

Certificate of Analysis (CoA) — Review Reference

Purpose: Field-by-field reference for reviewing a pharmaceutical/biopharma Certificate of Analysis. Original summary built from GMP release-testing practice (FDA 21 CFR 211.194, EU GMP, USP/Ph.Eur. monographs). Not a copy of any proprietary CoA.

Required header fields

- Product / material name and grade
- Manufacturer name and site
- Batch / lot number
- Manufacturing date and expiry (or retest) date
- CoA / document number and issue date
- Quantity / batch size
- Reference to the specification / monograph used (USP, Ph.Eur., in-house spec + version)

Test result table (per test)

- Test / attribute (e.g. appearance, identity, assay %, water content, pH, related substances, microbial limits, endotoxin, sterility where applicable)
- Specification / acceptance limit
- Method / procedure reference
- Result obtained
- Pass / Fail

Release statement

- Overall disposition: released / conforms to specification
- Name, title, signature of the authorized/qualified person
- Date of release

Review / practice checklist

- ☐ Batch/lot number matches the shipping record, the GDP record, and the physical container labels
- ☐ Every listed test has a result AND a stated acceptance limit (no blank result cells)
- ☐ Every result actually falls within its acceptance limit (recompute — don't trust the Pass column)
- ☐ Assay and impurity results are internally consistent (e.g. assay + impurities $\approx$ 100% where expected)
- ☐ Expiry/retest date present and consistent with the manufacturing date
- ☐ Specification version referenced and current
- ☐ CoA signed and dated by an authorized person
- ☐ Storage condition (if stated) matches the product label and the GDP record

Common findings

-  A result outside its acceptance limit marked “Pass” — Critical (out-of-spec release).
-  Sterility/endotoxin test missing on a sterile/injectable product — Critical.
-  Batch number does not match the shipment — Critical (wrong CoA for the lot).
-  Test listed with acceptance limit but no result — Major.
-  CoA unsigned / undated — Major.
-  Specification version not cited — Minor.

Sources

- FDA 21 CFR 211.194 (laboratory records): <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-C/part-211/subpart-J/section-211.194>
- USP General Notices and monographs: <https://www.usp.org/>
- ICH Q6A specifications guidance: <https://www.ich.org/page/quality-guidelines>